

The Ecosystem

Classification of companies shaping the semiconductor ecosystem



Fabless

Entities that design integrated circuits but strategically outsource manufacturing to external partners. This allows them to focus on innovation and design without the massive capital expenditure required for fabrication facilities—Qualcomm, NVIDIA, AMD

Integrated Device Manufacturers (IDMs)

Design, manufacture, assemble, test, and sell their semiconductor products across end markets. They control the entire vertical process, from chip conception to market delivery—Intel, Samsung, Micron

Foundries

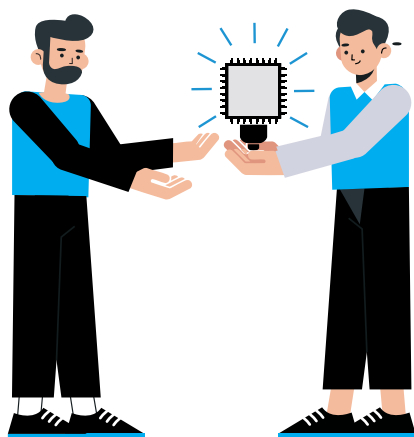
Specialise in wafer-level manufacturing or fabrication, which is often highlighted as the most capital-intensive step in the entire semiconductor value chain. Foundries provide manufacturing services for chip design houses like AMD, Qualcomm, Apple, and Nvidia—TSMC, GlobalFoundries, Intel Foundry Services, and Samsung Foundry

ATMP (Assembly, Testing, Marking, and Packaging)

Specialised facilities dedicated to the crucial final stages of chip production: assembly, testing, marking, and packaging. —Micron, Renesas

OSAT (Outsourced Semiconductor Assembly and Test)

External service providers that offer packaging services to various companies. —ASE Technology and Amkor Technology



GEOPOLITICS AT PLAY

Semiconductors is a borderless industry. Yet, some experts fear that US President Donald Trump's tariffs on India over Russian oil imports could cloud tech collaboration. "US-India relations could create uncertainty around technology transfer and joint ventures under the iCET framework. Geopolitical friction might slow investment flows or limit collaboration in sensitive areas. To mitigate this risk, India must diversify partnerships with Japan, Taiwan, South Korea, and Europe," says Helen Chiang, Semiconductor Research Lead and General Manager, IDC Taiwan. iCET is a collaborative framework established by the US and India to enhance cooperation in developing fields of technology. Just days after the Prime Minister's visit to Japan, Vaishnav confirmed that Rapidus Corporation, which is working on 2nm semiconductor technology, is keen on partnering with India.

The tariff debate also underscores how US' stance towards India differs from its posture towards China. "The term sanctions being used by the US administration on India has a very different cause of action than export control, tariffs and sanctions on China. Those are applied to chokehold the aggressive development of China's semiconductor ecosystem. In India's case, the US is not focusing on directly pausing the development of India's emerging semiconductor industry. In fact, India still enjoys the China plus one status. But the US is leveraging burgeoning exports, including electronics, out of India to derive geopolitical outcomes. The growth of the semiconductor industry may at some levels become collateral damage in the process," says Danish Faruqi, CEO of Fab Economics.

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DECISIVE JUNCTURE

India's bid to establish itself as a formidable semiconductor hub has transitioned from intent to execution. With ₹76,000 crore incentives already in play, and first-round funds nearly allocated, the industry is awaiting fresh investments.

The path ahead, particularly for Semicon 2.0, demands a strategic recalibration amid shifting geopolitical winds. Experts stress the next phase must address challenges flagged by chip makers and focus on strong supply chain clusters for chemicals, gases, wafers, and packaging materials. The government is already looking at incentives for downstream industries. Some experts say the government must show skin in the game to win the trust of global giants. Singapore has taken equity stakes in fabs, while the US is exploring similar models under the CHIPS Act. Washington recently bought a 10% stake in Intel.

The success of Tata Electronics' fab in Dholera will be India's make-or-break moment. Its outcome could determine whether global players expand into India and help it secure a seat at the world's most exclusive geopolitical table.

Conversely, failure risks reviving the mirages that have long haunted India's chip dreams. In semiconductors, delay is defeat, and the clock is ticking. **BT**